

DEPT: DECOUPLED EMBEDDINGS FOR
PRE-TRAINING LANGUAGE MODELS

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May 24, 2026

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Interference and inefficiency

- Negative interference: domains/languages compete for limited capacity.
- Curse of multilinguality: diminishing returns as languages increase, low-resource hurt most.
- Embedding bottleneck: shared giant vocabularies dilute tokens and dominate parameters (40–80%).

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Why heterogeneous pre-training

- Large language models power search, assistants, coding, and translation at global scale.
- Pre-training on diverse, massive corpora drives generalization and downstream success.
- Heterogeneous data (domains, languages) is essential but creates efficiency and stability challenges.

[illegible]

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Why existing practice struggles

- Heavy HPO: sampling temperatures per model-tokenizer pair; brittle stability [Arivazhagan et al., 2019].
- One-size-fits-all tokenizers: subword fragmentation and fertility across languages.
- Distributed training costs: embeddings dominate memory and communication in multilingual setups.

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- Train one transformer body across heterogeneous sources without a shared vocabulary.
- Keep token/positional embeddings local to each source; only aggregate the body.
- Federated/meta-learning style: local InnerOPT, global OuterOPT (FedAvg-like regularizer) [McMahan et al., 2017].



- GLOB: shared global embeddings; local training reduces comms by syncing every N_{local} steps.
- TRIM: per-source trimmed embeddings mapped to global; shrink local memory and comms.
- SPEC: fully decoupled token and positional embeddings per source; exchange only the body.

- Offline: build tokenizers (global or per-source), pre-tokenize each source.
- Inner loop: local training per source for a fixed number of local steps (AdamW; Loshchilov & Hutter, 2019).
- Outer loop: aggregate transformer bodies; embeddings handled per-variant.



- Body averaging injects stabilizing noise.
- Lowers activation and parameter norms.
- Enables stable training under fixed hyperparameters.



- TRIM: select source rows from the global embeddings; expand back by zero-padding before aggregation.
- SPEC: never aggregates token or position embeddings; only the transformer body is averaged.
- Positional specialization in SPEC supports unknown or private sequence characteristics.

- Savings increase with more local steps between aggregations.
- Larger gap between global and per-source vocabulary sizes yields larger savings.
- SPEC minimizes communication by not transmitting embeddings at all.

- SPEC-OPT cuts per-step communication by about 1600x at a 12-block multilingual setting.
- Embedding memory shrinks from 192M to 38.6M (about 5x) for SPEC-OPT.
- At 1B scale: 714x communication reduction; 24% model memory reduction.

See Table 2 below.

Table: Table 2: Practical memory and communication costs for DEPT

Type	# Blocks	Method	n_{local}	T	$V_b \times d_v$	$V_b \times d_{model}$	M_b (M)	Per-step Comm Cost (M)	
Multilingual	12	STD	5×103	1	250 112	192M	278 M (1x)	278 M (1x)	
Multilingual	12	GLOB	500	10	250 112	192M	278 M (1x)	0.26 M (0.000x)	
Multilingual	12	TRIM	500	10	216 135 ± 27 160	166M	252 M (0.90x)	0.5 M (0.002x)	
Multilingual	12	SPEC	500	10	216 135 ± 27 160	166M	252 M (0.90x)	0.17 M (0.0006x)	
Multilingual	12	SPEC-OPT	500	10	50 257 ± 0	38.6M	125 M (0.45x)	0.17 M (0.0006x)	
Multilingual (1B)	24	STD	7×103	1	250 112	512.2M	1.71 B (1x)	1.71 B (1x)	
Multilingual (1B)	24	SPEC-OPT	500	14	50 257 ± 0	102.9M	1.3B (0.76x)	2.4 M (0.001x)	

- Models: decoder-only transformers, 125M–1.3B parameters.
- Data: The Pile (16 subsets) and MC4 (8 langs), plus OOD splits.
- Metrics: validation perplexity (in/OOD), downstream tasks; stability via norms; plasticity via adaptation speed.

- Pile (24-block): about +15.3% average perplexity improvement; 10–20% OOD gains.
- MC4 (12-block): about +17.3% average in-distribution; up to +20.8% OOD.
- OuterOPT reduces norms and prevents divergence under heterogeneity.



- Average improvement of about 10–15% vs STD across subsets.
- SPEC-OPT is strongest on high UNIGRAM-CE subsets.
- Robust to embedding re-initialization across runs.

- Average in-distribution improvement of about 14–17%; OOD gains up to 20.8%.
- Consistent benefits across languages with varied tokenization difficulty.
- SPEC tends to be strongest on high-entropy OOD languages.

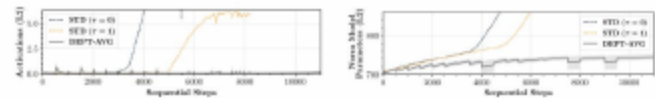
- DEPT improves MNLI, RACE, STSB, SST-2 by about 3–7.5% relative.
- TRIM is often strongest, suggesting better embedding–body synergy.
- Gains reflect stronger, more plastic transformer bodies.

Table: Table 7: Downstream task performance (↑) after continued pre-training

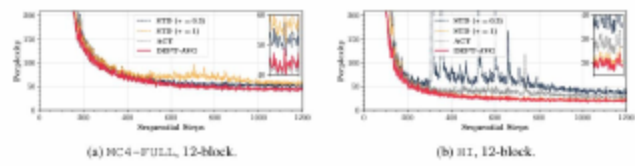
Name	Random Init RACE (ACC)	MNLI (ACC)	STSB (PC)	SST2 (ACC)						
STD ($\tau = 0$)	0.50	0.60	0.66	0.79						
STD ($\tau = 1$)	0.46	0.68	0.73	0.81						
GLOB	0.51	0.72	0.78	0.83						
TRIM	0.53	0.71	0.78	0.83						
SPEC	0.52	0.71	0.79	0.81						
SPEC-OPT	0.51	0.69	0.77	0.85						
Min Imp (%)	2.9%	4.6%	5.9%	-0.7%						
Max Imp (%)	5.8%	6.1%	7.5%	4.1%						

- DEPT adapts faster to new languages and domains from random embeddings.
- Reaches lower final perplexity on full distribution and OOD languages.
- SPEC's decoupled positions help transfer structural priors across sources.

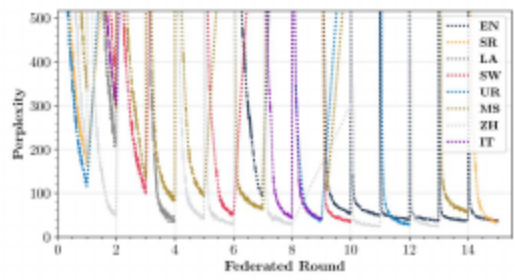
- First demonstration: SPEC converges stably with client subsampling.
- Late-introduced high-resource languages integrate without instability.
- Orders-of-magnitude comms reduction makes cross-silo training practical.



- Rapid convergence on full MC4, low-resource SW, and OOD (HI, DE).
- Lower perplexity floors indicate better cross-lingual transfer.
- Supports training with per-source optimized vocabularies (SPEC-OPT).



- On IID C4, DEPT matches standard perplexity aside from brief OuterOPT spikes.
- Maintains lower activations, suggesting training stability benefits.
- Indicates DEPT is not limited to heterogeneous settings.



- After SPEC training, build a shared embedding via short continued pre-training
- Alternatives: vocabulary matching or stitching; zero-shot mapping is open.
- This post-processing was applied uniformly across all methods.

- Zero-shot/global embedding construction without continued pre-training.
- Adaptive aggregation and client sampling; improved OOD alignment.
- Extend decoupling to more modalities; further scale federated pre-training.

- We thank collaborators and supporting institutions.
- Gratitude to dataset providers and open-source communities.

- Temperature-based multilingual sampling: Arivazhagan, Naveen, et al. 2019. Massively Multilingual Neural Machine Translation in the Wild.
- FedAvg (OuterOPT): McMahan, H. Brendan, et al. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data.
- AdamW (InnerOPT): Loshchilov, Ilya, and Frank Hutter. 2019. Decoupled Weight Decay Regularization.
- Implementation refs (optional): Kudo, T. 2018. SentencePiece; Press, O., et al. 2022. ALiBi.

- Can we build a global embedding without continued pre-training?
- What aggregation/client sampling best stabilizes small models?
- How to align OOD/private embeddings without leakage?
- Can decoupling extend to speech/vision tokenizers at scale?